

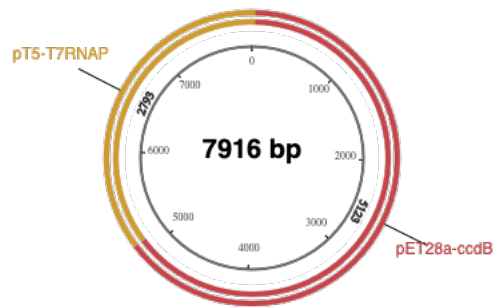
pET28-pT5-T7RNAP

Created: 4/10/2025, 9:37:51 AM

Saved: 4/10/2025, 10:22:02 AM

Component Fragments

Name	Length	Produced by	5' End	3' End
pET28a-ccdB	5123	PCR	Fwd Primer (auto)	Rev Primer (auto)
pT5-T7RNAP	2855	Synthetic	---	---



Notes

- Everything looks OK. No major issues detected.

Required oligos

Name	Primer 5' (overlap/spacer/ANNEAL) 3'	Len	%GC	3' %GC	3' Tm	3' Ta
pET28a-ccdB_fwd	GCTTGATCCGGCTGCTAAC	19	58	58	66.4	65.6
pET28a-ccdB_rev	ATTTCGCGGGATCGAGATC	19	53	53	64.6	65.6

Build Settings

Property	Value
Product/Kit	#E5520 NEBuilder HiFi DNA Assembly Cloning Kit
Minimum Overlap	20 nt
Minimum Overlap Tm	48 °C
Circularize	Yes
PCR Polymerase/Kit	Q5 High-Fidelity DNA Polymerase
PCR Primer Conc.	500 nM
Min. Primer Length	18 nt

Assembled Sequence

```
#LOCUS      pET28-pT5-T7RNAP      7916 bp ds-DNA circular SYN 10-APR-2025
#DEFINITION synthetic DNA
#ACCESSION  .
#VERSION    .
#KEYWORDS   NEBuilder
#SOURCE     synthetic DNA construct
# ORGANISM  synthetic DNA construct
#REFERENCE  1 (bases 1 to 7916)
# AUTHORS   .
# TITLE     NEBuilder-generated Construct
# JOURNAL    Exported 10-APR-2025 from NEBuilder https://nebuilder.neb.com
#COMMENT    NEBuilder-generated oligos (UPPERCASE = gene-specific, lowercase = overlap)
#COMMENT    pET28a-ccdB_fwd: GCTTGATCCGGCTGCTAAC
#COMMENT    pET28a-ccdB_fwd 3'Tm: 66.4 3'Ta: 65.6
#COMMENT    pET28a-ccdB_rev: ATTCGCGGGATCGAGATC
#COMMENT    pET28a-ccdB_rev 3'Tm: 64.6 3'Ta: 65.6
#FEATURES   Location/Qualifiers
#     source                1..7916
#                             /organism="synthetic DNA construct"
#                             /mol_type="other DNA"
#                             /plasmid="pET28-pT5-T7RNAP"
#     gene                  1..5123
#                             /note="pET28a-ccdB"
#     gene                    5124..30
#                             /note="pT5-T7RNAP"
#     primer_bind            1..19
#                             /note="pET28a-ccdB_fwd"
#                             /note="gene-specific Tm: 66.4 Ta: 65.6"
#                             /note="gene-specific primer: GCTTGATCCGGCTGCTAAC"
#     primer_bind            complement(5105..5123)
#                             /note="pET28a-ccdB_rev"
#                             /note="gene-specific Tm: 64.6 Ta: 65.6"
#                             /note="gene-specific primer: ATTCGCGGGATCGAGATC"
#ORIGIN
#       1 gcttgatccg gctgctaaca aagcccgaaa ggaagctgag ttggctgctg ccaccgctga
#      61 gcaataacta gcataacccc ttggggccctc taaacgggtc ttgaggggtt ttttgctgaa
#     121 aggaggaact atatccgatg ttggcgaatgg gacgcgccct gtagcggcgc attaacgcgc
#     181 gcgggtgtgg tggttacgcg cagcgtgacc gctacacttg ccagcgccct agcggccgct
#     241 cctttcgctt tcttcccttc ctttctcgcc acgttcgccg gctttccccg tcaagctcta
#     301 aatcgggggc tccctttagg gttccgattt agtgctttac ggcacctcga ccccaaaaaa
#     361 ctgattagg gtgatggttc acgtagtggg ccacgcgccct gatagacggt ttttcgccct
#     421 ttgacgttgg agtccacggt ctttaatatg ggactcttgt tccaaactgg aacaacactc
#     481 aaccctatct cggtctattc ttttgattta taagggattt tgccgatttc ggccatttgg
#     541 ttaaaaaaatg agctgattta acaaaaaattt aacgcgaatt ttaacaaaat attaacgttt
#     601 acaatttcag gtggcacttt tcgggggaaat gtgcgcggaa cccctatttg tttatttttc
#     661 taaatacatt caaatatgta tccgctcatg aattaattct tagaaaaact catcgagcat
#     721 caaatgaaac tgcaatttat tcatatcagg attatcaata ccatattttt gaaaaagccg
#     781 tttctgtaat gaaggagaaa actcaccgag gcagttccat aggatggcaa gatcctggta
#     841 tcggtctgcg attccgactc gtccaacatc aatacaacct attaatctcc cctcgtcaaa
#     901 aataaggtta tcaagtgaga aatcaccatg agtgacgact gaatccggtg agaatggcaa
#     961 aagtttatgc atttcttcc agacttggtc aacaggccag ccattacgct cgtcatcaaa
#    1021 atcactcgca tcaaccaaac cgttattcat tcgtgattgc gcctgagcga gacgaaatac
#    1081 gcgatcgtcg ttaaaaggac aattacaaac aggaatcgaa tgcaaccggc gcaggaacac
#    1141 tgccagcgca tcaacaatat ttccacctga atcaggatat tcttctaata cctggaatgc
#    1201 tgttttcccg gggatcgcat tggtagtaaa ccatgcatca tcaggagtac ggataaaatg
#    1261 ctgtaggtgc ggaagaggca taaattccgt cagccagttt agtctgacca tctcatctgt
#    1321 aacatcattg gcaacgctac ctttgccatg ttccagaaac aactctggcg catcgggctt
#    1381 cccatacaat cgatagattg tcgcacctga ttgcccgaca ttatcgcgag cccatttata
#    1441 cccataataa tcagcatcca tgttggaatt taatcgcggc ctagagcaag acgtttcccg
#    1501 ttgaatatgg tcataaacac cccttgattt actgtttatg taagcagaca gttttattgt
#    1561 tcatgaccaa aatcccttaa cgtgagtttt cgttccactg agcgtcagac cccgtagaaa
#    1621 agatcaaagg atcttcttga gatccttttt ttctgcgcgt aatctgctgc ttgcaaacia
#    1681 aaaaccacc gctaccagcg tgggtttggt tgccggatca agagctacca actcttttct
#    1741 cgaaggtaac tggcttcagc agagcgcaga taccaaaatac tgtccttcta gtgtagccgt
```

```

# 1801 agttaggcca ccacttcaag aactctgtag caccgcctac atacctcgct ctgctaatacc
# 1861 tgttaccagt ggctgtctgc agtggcgata agtcgtgtct taccgggttg gactcaagac
# 1921 gatagttacc ggataaggcg cagcggtcgg gctgaacggg gggttcgtgc acacagccca
# 1981 gcttggagcg accgacctac accgaactga gatacctaca gcgtgagcta tgagaaagcg
# 2041 ccacgcttcc cgaagggaga aaggcggaca ggtatccggg aagcggcagg gtcggaacag
# 2101 gagagcgcac gagggagctt ccagggggaa acgcctggta tctttatagt cctgtcgggt
# 2161 ttgccacct ctgacttgag cgtcgatttt tgtgatgtct gtcagggggg cggagcctat
# 2221 ggaaaaacgc cagcaacgcg gcctttttac ggttcctggc cttttgctgg ccttttgctc
# 2281 acatgttctt tcctgcgtta tccccgtatt ctgtggataa ccgtattacc gcctttgagt
# 2341 gagctgatac cgctcggcgc agccgaacga ccgagcgcag cgagtcagtg agcgaggaa
# 2401 cggagagcgc cctgatgcgg tattttctcc ttacgcatct gtgcggtatt tcacaccgca
# 2461 tatatggtgc actctcagta caatctgtct tgatgccga tagttaagcc agtatacact
# 2521 ccgctatcgc tacgtgactg ggtcatggct gcgccccgac acccgccaac acccgctgac
# 2581 gcgcccgtac gggcttgtct gctcccggca tccgcttaca gacaagctgt gaccgtctcc
# 2641 gggagctgca tgtgtcagag gttttcaccg tcacaccga aacgcgcgag gcagctgcgg
# 2701 taaagctcat cagcgtggtc gtgaagcgat tcacagatgt ctgcctgttc atccgcgtcc
# 2761 agctcgttga gttttccag aagcgttaat gtctggcttc tgataaagcg ggccatgtta
# 2821 agggcggttt tttcctgttt ggtcactgat gcctccgtgt aagggggatt tctgttcatt
# 2881 ggggtaatga taccgatgaa acgagagagg atgctcagca tacgggttac tgatgatgaa
# 2941 catgcccggt tactggaacg ttgtgagggg aaacaactgg cggtatggat gcggcgggac
# 3001 cagagaaaaa tcactcaggg tcaatgccag cgcttcgtta atacagatgt aggtgttcca
# 3061 cagggtagcc agcagcatcc tgcgatgcag atccggaaca taatggtgca gggcgtgac
# 3121 ttccgcgttt ccagacttta cgaaacacgg aaaccgaaga ccattcatgt tgttgcacg
# 3181 gtgcagacg ttttgacga gcagtcgctt cacgttcgct cgcgtatcgg tgattcattc
# 3241 tgctaaccag taaggcaacc ccgccagcct agccgggtcc tcaacgacag gacacgac
# 3301 atgcgcaccc gtggggccgc catgcccggc ataatggcct gcttctcgcc gaaacgtttg
# 3361 gtggcgggac cagtgcagaa ggcttgagcg agggcgtgca agattccgaa taccgcaagc
# 3421 gacaggcgca tcctgtcgc gctccagcga aagcgtccct cgcgaaaaat gaccagagc
# 3481 gctgcccggc cctgtcctac gagttgcatg ataaagaaga cagtcataag tgcgcgacg
# 3541 atagtcatgc ccgcgcacca ccggaaggag ctgactgggt tgaaggctct caagggcac
# 3601 ggtcgagatc ccggtgccta atgagtgcgc taacttacat taattgcgtt gcgctcactg
# 3661 cccgctttcc agtcgggaaa cctgtcgtgc cagctgcatt aatgaatcgg ccaacgcgcg
# 3721 gggagaggcg gtttgctgat tgggcgccag ggtgggtttt cttttacca gtgagacggg
# 3781 caacagctga ttgcccttca ccgcctggcc ctgagagagt tgcagcaagc ggtccacgct
# 3841 ggtttgcccc agcaggcgaa aatcctgttt gatggtggtt aacggcggga tataacatga
# 3901 gctgtcttgc gtatcgtcgt atcccactac cgagatatcc gcaccaacgc gcagcccgga
# 3961 ctcggtaatg gcgcgcattg cgcccagcgc catctgatcg ttggcaacca gcctgcagt
# 4021 gggaacgatg cccctattca gcatttgcag ggtttgttga aaaccggaca tggcactcca
# 4081 gtgccttcc cgttccgcta tcggctgaat ttgattgcga gtgagatatt tatgccagcc
# 4141 agccagacgc agacgcgccg agacagaact taatgggccc gctaacagcg cgatttgctg
# 4201 gtgacccaat gcgaccagat gctccacgcc cagtcgcgta ccgtcttcat gggagaaaat
# 4261 aatactgttg atgggtgtct ggtcagagac atcaagaaat aacgccggaa cattagtga
# 4321 ggcagcttcc acagcaatgg catcctggtc atccagcgga tagttaatga tcagccact
# 4381 gacgcgttgc gcgagaagat tgtgcaccgc cgctttacag gcttcgacgc cgcttcgttc
# 4441 taccatcgac accaccacgc tggcaccag ttgatcggcg cgagatttaa tcgcccgcg
# 4501 aatttgcgac gcgcggtgca gggccagact ggagggtgca acgccaatca gcaacgactg
# 4561 tttgcccgcc agttgttgtg ccacgcgggtt gggaatgtaa ttcagctccg ccatcgccgc
# 4621 ttccactttt tcccgcgttt tcgcagaaac gtggctggcc tggttacca cgcggaac
# 4681 ggtctgataa gagacaccgg catactctgc gacatcgtat aacgttactg gttcacatt
# 4741 caccaccctg aattgactct cttccgggcg ctatcatgcc ataccgcgaa aggttttgcg
# 4801 ccattcgatg gtgtccggga tctcgacgct ctcccttatg cgactcctgc attaggaagc
# 4861 agcccagtag taggttgagg ccgttgagca ccgcgcgcgc aaggaatggt gcatgcaagg
# 4921 agatggcgcc caacagtccc ccggccacgg ggcctgccac cataccacg ccgaaacaag
# 4981 cgctcatgag ccgaagtgg cgagcccgat cttcccctac ggtgatgtcg gcgatatagg
# 5041 cgccagcaac cgcacctgtg gcgcccgtga tgcggccac gatgcgtccg gcgtagagga
# 5101 tcgagatctc gatcccgcga aattcataaa aaatttattt gctttgtgag cggataacaa
# 5161 ttataataga ttcaattgtg agcggataac aatttcacac agaattcatt aaagaggaga
# 5221 aattaactat gagaggatcg catcaccatc accatcacgg atccatgaac acgattaaca
# 5281 tcgctaagaa cgacttctct gacatcgaac tggctgctat cccgttcaac actctggctg
# 5341 accattacgg tgagcgttta gctcgcgaac agttggccct tgagcatgag tcttacgaga
# 5401 tgggtgaagc acgcttccgc aagatgtttg agcgtcaact taaagctggt gaggttgcgg
# 5461 ataacgctgc gcgcaagcct ctcatacta cctactccc taagatgatt gcacgcatca
# 5521 acgactggtt tgaggaaagt aaagctaagc gcggcaagcg cccgacagcc ttccagttcc
# 5581 tgcaagaaat caagccggaa gccgtagcgt acatcaccat taagaccact ctggcttgcc
# 5641 taaccagtgc tgacaatata accgttcagg ctgtagcaag cgcaatcggg cgggccattg
# 5701 aggacgagcg tcgcttcggt cgtatccgtg acctgaagc taagcacttc aagaaaaacg
# 5761 ttgagggaaca actcaacaag cgcgtagggc acgtctacaa gaaagcattt atgcaagtgt
# 5821 tcgaggctga catgctctct aagggtctac tcggtggcga ggcgtggtct tctggcata
# 5881 aggaagactc tattcatgta ggagtagcgt gcacgcagat gctcattgag tcaaccggaa

```

```

# 5941 tggtagctt acaccgcaa aatgctggcg tagtaggtca agactctgag actatcgaac
# 6001 tcgcacctga atacgctgag gctatcgcaa cccgtgcagg tgcgctggct ggcatctctc
# 6061 cgatgttcca accttgcgta gttcctccta agccgtggac tggcattact ggtggtggct
# 6121 attgggctaa cggtcgtcgt cctctggcgc tggcgctac tcacagtaag aaagcactga
# 6181 tgcgctacga agacgtttac atgcctgagg tgtacaaagc gattaacatt gcgcaaaaca
# 6241 ccgcatggaa aatcaacaag aaagtcctag cggtcgcaa cgtaatcacc aagtgaagc
# 6301 attgtccggt cgaggacatc cctgcgattg agcgtgaaga actcccgatg aaaccggaag
# 6361 acatcgacat gaatcctgag gctctcaccg cgtggaaacg tgctgccgct gctgtgtacc
# 6421 gcaaggacaa ggctcgcaag tctcgccgta tcagccttga gttcatgctt gagcaagcca
# 6481 ataagtttgc taaccataag gccatctggt tcccttaca catggactgg cgcggtcgtg
# 6541 tttagctgt gtcaatgttc aaccgcaag gtaacgatat gaccaaagga ctgcttacgc
# 6601 tggcgaaagg taaaccaatc ggtaaggag gttactactg gctgaaaatc cacggtgcaa
# 6661 actgtgctgg tgctgataag gttccgttcc ctgagcgcat caagttcatt gaggaaaacc
# 6721 acgagaacat catggcttgc gctaagtctc cactggagaa cacttggtgg gctgagcaag
# 6781 attctccgtt ctgcttccct cgttctgct ttgagtacgc tggggtacag caccacggcc
# 6841 tgagctataa ctgctccctt ccgctggcgt ttgacgggtc ttgctctggc atccagcact
# 6901 tctcccgcat gctccgagat gaggtagggt gtcgcgcggt taacttgctt cctagtgaag
# 6961 ccgttcagga catctacggg attgttgcta agaaagtcaa cgagattcta caagcagacg
# 7021 caatcaatgg gaccgataac gaagtagtta ccgtgaccga tgagaacact ggtgaaatct
# 7081 ctgagaaagt caagctgggc actaaggcac tggctggtca atggctggct tacggtgtta
# 7141 ctgcgagtgt gactaagcgt tcagtcatga cgctggctta cgggtccaaa gatttcggct
# 7201 tccgtcaaca agtgcaggaa gataccattc agccagctat tgattccggc aagggtctga
# 7261 tgttactca gccgaatcag gctgctggat acatggctaa gctgatttgg gaatctgtga
# 7321 gcgtgacggt ggtagctgcg gttgaagcaa tgaactggct taagtctgct gctaagctgc
# 7381 tggctgtgta ggtcaaagat aagaagactg gagagattct tcgcaagcgt tgcgctgtgc
# 7441 attgggtaac tcctgatggt ttccctgtgt ggcaggaata caagaagcct attcagacgc
# 7501 gcttgaacct gatgttccct ggctagttcc gcttacagcc taccattaac accaacaag
# 7561 atagcgagat tgatgcacac aaacaggagt ctggtatcgc tcctaacttt gtacacagcc
# 7621 aagacgtag ccaccttcgt aagactgtag tgtgggcaca cgagaagtac ggaatcgaat
# 7681 cttttgact gattcacgac tccttcggta ccattccggc tgacgctgcg aacctgttca
# 7741 aagcagtgcg cgaaactatg gttgacacat atgagtcttg tgatgtactg gctgatttct
# 7801 acgaccagtt cgctgaccag ttgcacgagt ctcaattgga caaaatgcca gcacttccgg
# 7861 ctaaaggtaa cttgaacctc cgtgacatct tagagtcgga cttcgcgttc gcgtaa

```

```

#//
#

```